

Structural and Angiographic Assessment of the Nerve Fiber Layer in Age-Related Macular Degeneration and Diabetic Retinopathy

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PURPOSE. The purpose of this study was to evaluate structural and microvascular alterations in the nerve fiber layer (NFL) in age-related macular degeneration (AMD) and diabetic retinopathy (DR).

METHODS. A total of 74 healthy controls, 101 patients with AMD, and 117 patients with DR were included. One eye per participant was imaged using a Solix OCTA device centered on the optic disc to measure NFL thickness and nerve fiber layer plexus (NFLP) vessel density (VD). For both parameters, we analyzed the independent associations of age, AMD, and DR; compared age-matched healthy controls with the AMD groups and with the DR groups; and evaluated correlations with age in healthy controls.

RESULTS. Multivariate analyses demonstrated that age was independently associated with both NFL thickness and NFLP VD, whereas AMD was not associated with either parameter and DR was associated only with NFLP VD. The associations of AMD and DR with these two parameters were also confirmed in age-matched analyses. Additionally, NFLP VD decreased significantly with increasing DR severity ($P < 0.001$). In healthy controls, age was significantly correlated with both NFL thickness (Spearman's $\rho = -0.26$, $P = 0.026$) and NFLP VD ($\rho = -0.32$, $P = 0.006$).

CONCLUSIONS. NFL thickness was associated with age, but not with AMD or DR. In contrast, DR was independently associated with reduced NFLP VD, which also significantly correlated with DR severity. These findings provide evidence supporting further investigation of NFLP VD in relation to DR, while accounting for age.

Keywords: OCT, OCTA, AMD, DR, NFL

Age-related macular degeneration (AMD) and diabetic retinopathy (DR) are leading causes of severe and irreversible vision loss.^{1,2} Although AMD mainly affects the outer retinal layers and DR has traditionally been regarded as a microvascular disorder, accumulating evidence indicates that both conditions involve progressive inner retinal neurodegeneration.^{3–5} The retinal nerve fiber layer (NFL), which forms the innermost neural layer of the retina and consists of unmyelinated axons of retinal ganglion cells, relies on the nerve fiber layer plexus (NFLP), which corresponds to the radial peripapillary capillary plexus, for metabolic support.⁶ Neurodegenerative changes and microvascular impairments can induce structural alterations in the NFL and compromise capillary integrity, potentially leading to visual field defects. Assessing both structural and microvascular variation within the NFL may enhance monitoring of disease progression in AMD and DR, facilitating early detection and clinical management.

Optical coherence tomography (OCT) is a noninvasive, high-resolution imaging modality that provides depth-resolved images of the retina. It is commonly used to

assess NFL structure.^{7–9} However, standard OCT has difficulty measuring microvasculature. OCT angiography (OCTA) enhances traditional OCT by incorporating vascular imaging at the capillary level, achieved through measuring motion contrast between successive scans.¹⁰ OCTA retains advantages from standard OCT imaging, including noninvasive volumetric data acquisition,^{11–13} and it has demonstrated utility in identifying structural and microvascular alterations in retinopathies.^{14,15} Both OCT and OCTA data can be acquired simultaneously during a single scan. OCT volumes are used for structural evaluation, revealing tissue changes caused by various ocular disorders, whereas OCTA volumes are used to assess microvascular changes.

Although several studies have investigated changes in NFL thickness or NFLP vessel density (VD) in AMD and DR,^{8,16} the impact of potential confounding factors on these parameters remains poorly understood. Clarifying how such factors influence NFL thickness and NFLP VD can improve our understanding of their relationships with AMD and DR. In this study, we used OCTA to simultaneously obtain structural and angiographic data, allowing the quantification of

NFL thickness and NFLP VD. We investigated the associations of these parameters with AMD, DR, and age, a potential confounding factor, to explore their relationship with these diseases.

METHODS

Study Population

All participants in this study were recruited at the Casey Eye Institute, Oregon Health and Science University, and obtained informed written consent. This research was approved by the institutional review board of Oregon Health and Science University, and all procedures were followed according to protocol in accordance with the Declaration of Helsinki.

Each participant underwent a comprehensive clinical examination at the Casey Eye Institute and was classified as healthy or diagnosed with either AMD or DR by two retina specialists (authors S.T.B. and T.S.H.). Healthy controls showed no clinical evidence of AMD, DR, or other retinal pathology identified during ophthalmic examination and retinal imaging. Patients with AMD were further classified into early AMD, intermediate AMD, and advanced AMD (geographic atrophy [GA] or neovascular AMD) according to the Age-Related Eye Disease Study (AREDS) classification.¹⁷ Patients with DR were further classified into mild to moderate nonproliferative diabetic retinopathy (NPDR), severe NPDR, and proliferative diabetic retinopathy (PDR) based on the Early Treatment of Diabetic Retinopathy Study (ETDRS) protocol.

Participants were excluded if they met any of the following criteria: (1) a history of intraocular surgery other than uncomplicated cataract extraction with posterior chamber intraocular lens implantation; (2) the presence of any ocular or systemic disease other than AMD or DR that could affect the optic disc, including glaucoma, optic neuritis, ischemic optic neuropathy, papilledema, and optic nerve atrophy; (3) a history of panretinal photocoagulation; (4) an inability to cooperate or provide reliable imaging data; and (5) OCT/OCTA images with low signal strength index (SSI < 50), significant motion artifacts, defocusing, or off-center positioning were excluded from the study.

Data Acquisition and Processing

A commercial OCTA system (Solix, Visionix/Optovue Inc., Fremont, CA, USA) was used to scan all participants in a 6 × 6-mm area centered on the optic disc. The device worked with a center wavelength of 840 nm and 120-kilohertz (kHz) axial scan rate. Each scan volume consisted of 512 line-scan locations, 2 repeated B-scans were acquired at each location, and each B-scan contained 512 A-lines. OCTA flow signal was detected using the commercial version of the split-spectrum amplitude-decorrelation angiography (SSADA) algorithm.¹⁰

The inner limiting membrane (ILM) and inner boundary of the ganglion cell layer (GCL) were segmented using a graph search-based algorithm.¹⁸ The segmentation errors were corrected using the Center for Ophthalmic Optics & Lasers-Angiography Reading Toolkit (COOL-ART) software, which incorporates an intelligent manual correction module.¹⁹ All layer segmentations were subsequently reviewed and corrected by clinicians, and no vitreoretinal traction-related abnormalities affecting the measurements

were identified. En face NFL thickness maps were generated by measuring the distance from the ILM to the inner boundary of the GCL. The NFL thickness values used for statistical analysis were obtained by averaging the measurements across the 6 × 6-mm map, excluding a central 2-mm-diameter circle manually centered on the optic disc based on the en face reflectance image. En face NFLP angiograms were generated from the OCTA volumes using maximum projection within the NFL slab.²⁰ In the en face OCTA image of the NFLP, retinal vasculature was enhanced using local contrast, and then Otsu's method was applied to generate a flow signal binary vascular mask.²¹ The NFLP VD was measured as the percentage area with flow signal within the same region used for NFL thickness analysis. All OCT and OCTA images were adjusted for axial length-related linear magnification prior to quantitative analysis, as described previously.^{22,23}

Statistical Methods

All statistical analyses were conducted using SPSS, version 29.0 (IBM Corp., Armonk, NY, USA). A multivariate linear regression analysis was performed to evaluate the factors associated with NFL thickness and NFLP VD, including participants' age, AMD, and DR. To minimize the potential confounding effect of age, age-adjusted subgroup analyses were performed by restricting the study population to cohorts with comparable age distributions. Age-matched comparisons analyses were then repeated using the age-adjusted cohorts (Supplementary Tables S1, S2). The Shapiro-Wilk test was used to assess normality. Because some variables were not normally distributed, the nonparametric Kruskal-Wallis H test was used for comparisons among more than two groups. Following a significant Kruskal-Wallis test, post hoc pairwise comparisons between groups were performed using Dunn's tests with Bonferroni correction to adjust for multiple comparisons and control the family-wise error rate. Spearman rank-order correlation analysis was used to test the correlation between NFL thickness/NFLP VD and participants' age or disease severity.

RESULTS

Study Population

A total of 74 healthy controls, 101 patients with AMD, and 117 patients with DR were included in this study (Table 1). Only one eye from each participant was analyzed. The AMD group comprised 52 patients with early to intermediate AMD and 49 with advanced AMD (GA or neovascular). The DR group comprised 63 patients with mild to moderate NPDR, 34 with severe NPDR, and 20 with PDR.

Multivariate Analyses of NFL Thickness and NFLP VD in AMD and DR Eyes

In the multivariate analyses for eyes with AMD, age was significantly associated with reductions in NFL thickness ($P = 0.002$) and NFLP VD ($P < 0.001$). AMD was not significantly associated with either NFL thickness or NFLP VD. Therefore, age emerged as the primary factor associated with reductions in NFL thickness and NFLP VD in patients with AMD (Table 2).

Multivariate analyses for eyes with DR revealed that age was significantly associated with reduced NFL thickness

TABLE 1. Characteristics of Participants

Parameter	Healthy Controls	AMD		DR		
		Early to Intermediate AMD	Advanced AMD	Mild to Moderate NPDR	Severe NPDR	PDR
Eyes, <i>n</i>	74	52	49	63	34	20
Gender, M/F	30/44	24/28	15/34	34/29	19/15	9/11
Age, mean $\pm$ SD, y	54.49 $\pm$ 15.27	74.15 $\pm$ 6.47	75.69 $\pm$ 6.17	62.59 $\pm$ 12.31	61.41 $\pm$ 9.81	46.55 $\pm$ 10.59
Age range, y	28–79	55–85	58–86	23–88	36–75	31–66
Type 1/type 2 diabetes, <i>n</i>	NA	NA	NA	19/44	4/30	11/9
Anti-VEGF injection, <i>n</i> (%)	VA	NA	29 (59)	19 (30)	18 (53)	7 (35)
Diabetes duration, y	NA	NA	NA	20.75 $\pm$ 12.07	15.21 $\pm$ 10.69	20.66 $\pm$ 10.31
Axial length, mm	23.85 $\pm$ 0.87	24.16 $\pm$ 0.99	23.14 $\pm$ 1.22	23.83 $\pm$ 1.09	23.52 $\pm$ 0.72	23.39 $\pm$ 0.92
SSI	73.72 $\pm$ 8.14	65.91 $\pm$ 5.69	68.29 $\pm$ 6.33	65.79 $\pm$ 9.33	66.21 $\pm$ 6.56	71.3 $\pm$ 10.25
BCVA (LogMAR)	0.03 $\pm$ 0.14	0.13 $\pm$ 0.14	0.14 $\pm$ 0.16	0.13 $\pm$ 0.17	0.16 $\pm$ 0.21	0.09 $\pm$ 0.17
NFL thickness, mean $\pm$ SD, μ m	77.81 $\pm$ 8.01	77.32 $\pm$ 10.54	74.61 $\pm$ 11.24	76.07 $\pm$ 10.46	76.38 $\pm$ 15.14	77.75 $\pm$ 13.23
NFLP VD, mean $\pm$ SD	0.57 $\pm$ 0.04	0.53 $\pm$ 0.05	0.52 $\pm$ 0.06	0.54 $\pm$ 0.05	0.52 $\pm$ 0.05	0.55 $\pm$ 0.05

AMD, age-related macular degeneration; BCVA, best-corrected visual acuity; DR, diabetic retinopathy; F, Female; M, Male; NA, not applicable; NFL, nerve fiber layer; NFLP VD, nerve fiber layer plexus vessel density; NPDR, non-proliferative diabetic retinopathy; PDR, proliferative diabetic retinopathy; SD, standard deviation; SSI, Signal Strength Index.

TABLE 2. Multivariate Analyses of NFL Thickness and NFLP VD for Eyes With AMD

Parameter	Age		AMD	
	Standard β	<i>P</i> Value	Standard β	<i>P</i> Value
NFL thickness	−0.31	0.002	0.12	0.23
NFLP VD	−0.4	<0.001	−0.17	0.055

The *P* values in bold represent statistical significance.

TABLE 3. Multivariate Analyses of NFL Thickness and NFLP VD for Eyes With DR

Parameter	Age		DR	
	Standard β	<i>P</i> Value	Standard β	<i>P</i> Value
NFL thickness	−0.25	<0.001	−0.004	0.96
NFLP VD	−0.42	<0.001	−0.26	<0.001

The *P* values in bold represent statistical significance.

($P < 0.001$), whereas the presence of DR was not significantly associated with NFL thickness. Both age and DR were significantly associated with reduced NFLP VD (all $P < 0.001$; [Table 3](#); Supplementary Fig. S1).

Therefore, in multivariate analysis, age was a significant confounding factor associated with NFL thickness and NFLP VD in eyes with either AMD or DR. The presence of AMD was not significantly associated with either parameter. In contrast, the presence of DR was significantly associated with NFLP VD but not with NFL thickness. We further investigated differences in NFL thickness and NFLP VD between age-matched groups.

Comparison Between Age-Matched Healthy Controls and Patients With AMD

Given that age was significantly associated with NFL thickness and NFLP VD, age matching was performed between the healthy control group and the AMD groups prior to group comparisons (see Supplementary Table S1). Using the

nonparametric Kruskal-Wallis H test, no statistically significant difference was observed in NFL thickness among healthy control group (mean $\pm$ SD = 75.6 $\pm$ 7.53 μ m), the early to intermediate AMD group (78.08 $\pm$ 9.54 μ m), and the advanced AMD group (79.8 $\pm$ 10.82 μ m). Similarly, no statistically significant difference was observed in NFLP VD among the healthy control group (0.56 $\pm$ 0.04), the early to intermediate AMD group (0.55 $\pm$ 0.05), and the advanced AMD group (0.54 $\pm$ 0.05).

Comparison Between Age-Matched Healthy Controls and Patients With DR

Comparisons of NFL thickness and NFLP VD between the age-matched healthy controls and DR groups (see Supplementary Table S2) were performed using the nonparametric Kruskal-Wallis H test. No significant differences in NFL thickness were observed, whereas significant differences in NFLP VD were observed among the groups ([Table 4](#)). We further performed Dunn's test with Bonferroni correction for post hoc pairwise comparisons of NFLP VD ([Fig. 1](#)). The NFLP VD was lower in the DR severity groups than in the healthy control group. Among these comparisons, NFLP VD was significantly lower in the severe NPDR and PDR groups compared with the healthy control group ($P < 0.001$ and $P = 0.007$, respectively).

Correlation of NFL Thickness and NFLP VD With DR Severity and Participant Age

The Spearman correlation coefficients for NFL thickness and NFLP VD revealed that the NFL thickness was significantly correlated with participant age (Spearman's $\rho = -0.18$, $P = 0.03$) but not with DR severity ($\rho = -0.04$, $P = 0.66$). The NFLP VD correlated with DR severity ($\rho = -0.32$, $P < 0.001$) and participant age ($\rho = -0.33$, $P < 0.001$). Further visual inspection of OCT and OCTA images from healthy controls and different DR severity groups was performed ([Fig. 2](#)), with participants of similar age. No notable changes in en face NFL thickness images were observed (see [Fig. 2](#), row 4). However, we observed vessel dropout with DR

TABLE 4. Comparison of NFL Thickness and NFLP VD Between Age-Matched Healthy Controls and DR Groups

Subject	Healthy Controls	Mild to Moderate NPDR	Severe NPDR	PDR	Difference, <i>P</i> Value*
NFL thickness, mean $\pm$ SD, μ m	77.83 $\pm$ 8.24	77.59 $\pm$ 11.16	79.39 $\pm$ 14.38	76.57 $\pm$ 13.91	0.75
NFLP VD, mean $\pm$ SD	0.57 $\pm$ 0.04	0.56 $\pm$ 0.05	0.53 $\pm$ 0.05	0.54 $\pm$ 0.05	0.001

* Kruskal-Wallis H-test, *P* value < 0.05 was considered significant. The *P* values in bold represent statistical significance.

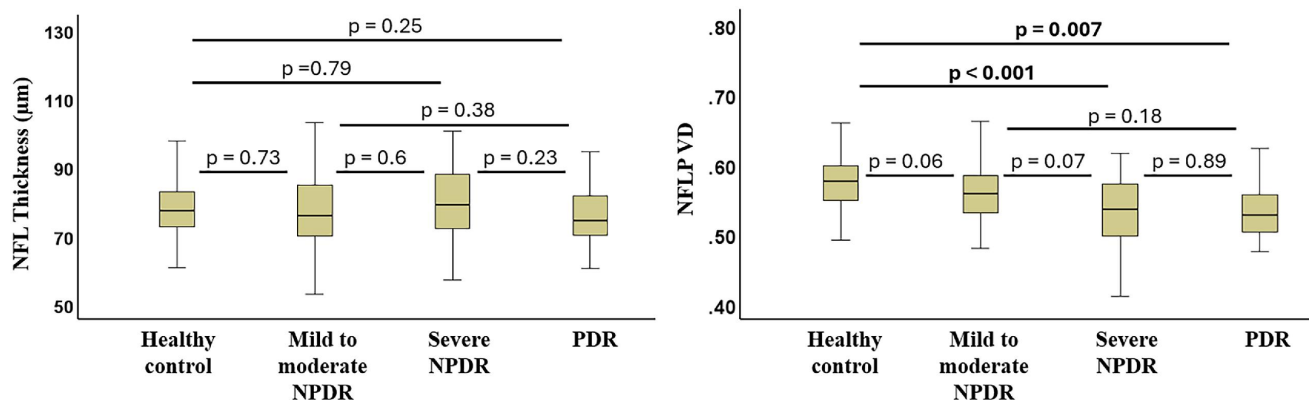


FIGURE 1. Boxplots of nerve fiber layer (NFL) thickness (**left**) and NFL plexus (NFLP) vessel density (VD) (**right**). Post hoc pairwise comparisons between groups were performed using Dunn's test with Bonferroni correction. A *P* value < 0.008 (0.05/6, accounting for 6 pairwise comparisons among 4 groups) was considered statistically significant. Significant comparisons are indicated in bold. NPDR, nonproliferative diabetic retinopathy; PDR, proliferative diabetic retinopathy.

progression on NFLP angiograms and NFLP VD maps. The NFLP VD map displayed a more pronounced reduction in vessel density as DR severity increased (see Fig. 2, rows 2 and 3).

Correlation of Age With NFL Thickness and NFLP VD in the Healthy Controls

In the healthy control group, we evaluated the Spearman correlation coefficient of NFL thickness and NFLP VD with age. A significant negative correlation was observed between age and both NFL thickness and NFLP VD ($\rho = -0.26$, $P = 0.026$ and $\rho = -0.32$, $P = 0.006$, respectively; Fig. 3). We visually inspected the changes in NFL thickness and NFLP VD from younger to older participants to further validate these findings. All healthy controls were divided into 6 age-based subgroups: 21 to 30, 31 to 40, 41 to 50, 51 to 60, 61 to 70, and 71 to 80 (Fig. 4). No structural abnormalities were observed in the cross-sectional OCT images. Although changes in NFL thickness were subtle, a decrease with increasing age could still be observed on the NFL thickness maps (see Fig. 4, row 4). Meanwhile, vascular attenuation was evident in the NFLP angiograms, and a reduction in VD was observed in the NFLP VD maps (see Fig. 4, rows 2 and 3).

DISCUSSION

This study investigated factors associated with reductions in NFL thickness and NFLP VD using OCTA and explored their relationship with AMD and DR. We evaluated the associations of age and AMD and DR with both NFL thickness and NFLP VD. Our findings demonstrate the follow-

ing: (1) in multivariate analyses, age was identified as a significant confounding factor for NFL thickness and NFLP VD, AMD showed no significant association with these parameters, whereas DR was significantly associated with NFLP VD but not with NFL thickness; (2) no significant differences were observed in NFL thickness and NFLP VD between age-matched healthy controls and patients with AMD; (3) significant differences in NFLP VD were observed between age-matched healthy controls and patients with DR, whereas no difference in NFL thickness were observed; moreover, NFLP VD was correlated with DR severity and age, whereas NFL thickness was correlated only with age; and (4) significant negative correlations with age were found for both NFL thickness and NFLP VD in the healthy control group.

Notably, age plays an important role in reductions of both NFL thickness and NFLP VD. Multivariate analysis revealed that age is significantly associated with alterations in NFL thickness and NFLP VD in patients with AMD and DR (see Tables 2, 3). In healthy controls, both parameters showed clear negative correlations with increasing age (see Fig. 3), which could also be visualized in NFL thickness and NFLP VD maps (see Fig. 4, rows 3 and 4). Our observation of decreasing NFL thickness with age is consistent with previous reports showing an inverse relationship between RNFL thickness and age.^{24,25} Similarly, the decline in NFLP VD with age aligns with earlier studies demonstrating a significant association between reduced NFLP VD and older age.^{26,27} Overall, our findings indicate that age is strongly associated with reductions in NFL thickness and NFLP VD across healthy individuals as well as patients with AMD and DR. These results underscore why age should be carefully matched or controlled when analyzing NFL thickness and NFLP VD.

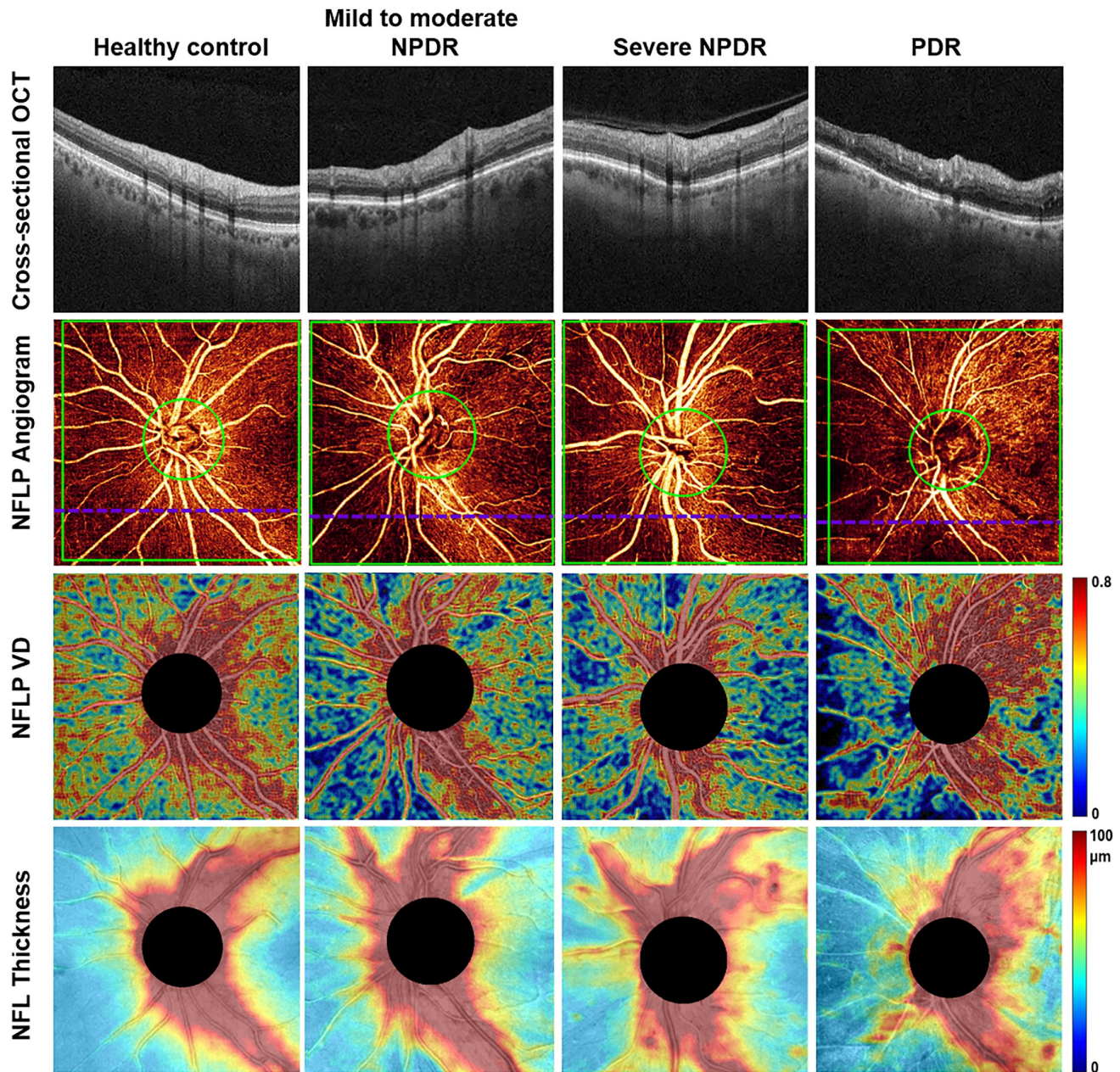


FIGURE 2. Representative OCT and OCTA images from a healthy control group and DR groups (mild to moderate NPDR, severe NPDR, and PDR). The cross-sectional OCT images (row 1) are shown with the corresponding location indicated by the *violet dashed line*. The NFLP angiogram (row 2) is overlaid with the measurement area, defined by the *green square* while excluding a 2-mm-diameter *circle* centered on the optic disc. NFLP VD (row 3) decreases with increasing DR severity, while NFL thickness (row 4) remains relatively stable across different DR severities. Participant age difference is no more than 3 years. OCT, optical coherence tomography; OCTA, optical coherence tomography angiography; NFLP VD, nerve fiber layer plexus vessel density; NFL, nerve fiber layer; NPDR, nonproliferative diabetic retinopathy; PDR, proliferative diabetic retinopathy.

NFL thinning and reductions in NFLP VD were not associated with AMD in this study. Multivariate analysis revealed that AMD was not associated with NFL thickness or NFLP VD. We further investigated age-matched healthy controls and the AMD groups on these parameters, and no significant differences in NFL thickness or NFLP VD were observed. These findings on NFL thickness are consistent with previous reports showing no statistically significant difference in NFL thickness between patients with

early-stage AMD and individuals without AMD.²⁸ Additionally, we provide new insights, demonstrating that NFLP VD does not differ between healthy controls and AMD groups. Together, these results indicate that NFL thickness and NFLP VD have limited utility as biomarkers for assessing patients with AMD.

The presence of DR was significantly associated with reduced NFLP VD but not with reduced NFL thickness. Multivariate analysis revealed a strong association between DR

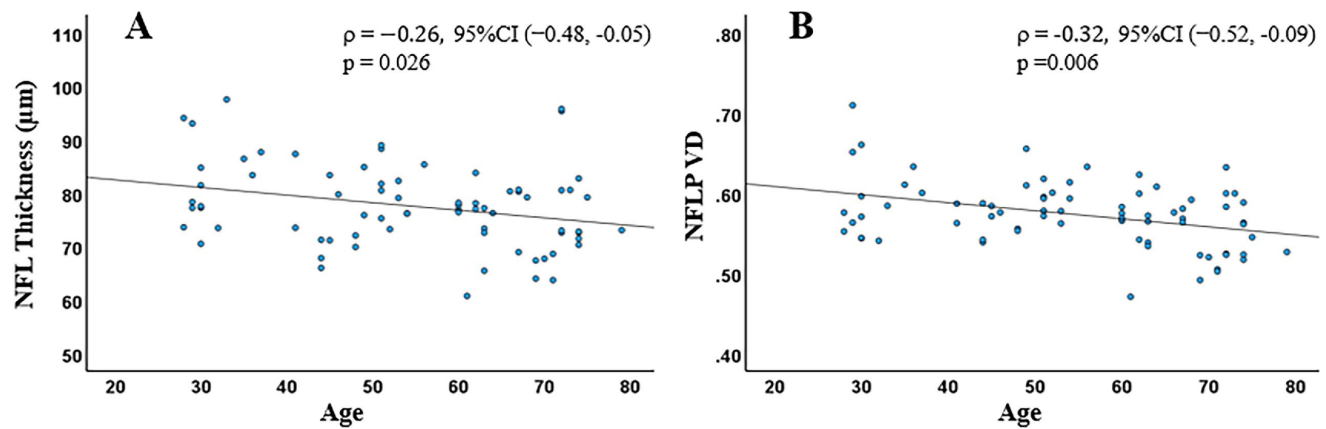


FIGURE 3. The relationship between age and nerve fiber layer (NFL) thickness (**A**) and nerve fiber layer plexus (NFLP) vessel density (VD) (**B**) in healthy controls. Spearman's ρ and P values were calculated using Spearman rank correlation. NFL thickness and NFLP VD were significantly correlated with age. CI, confidence interval.

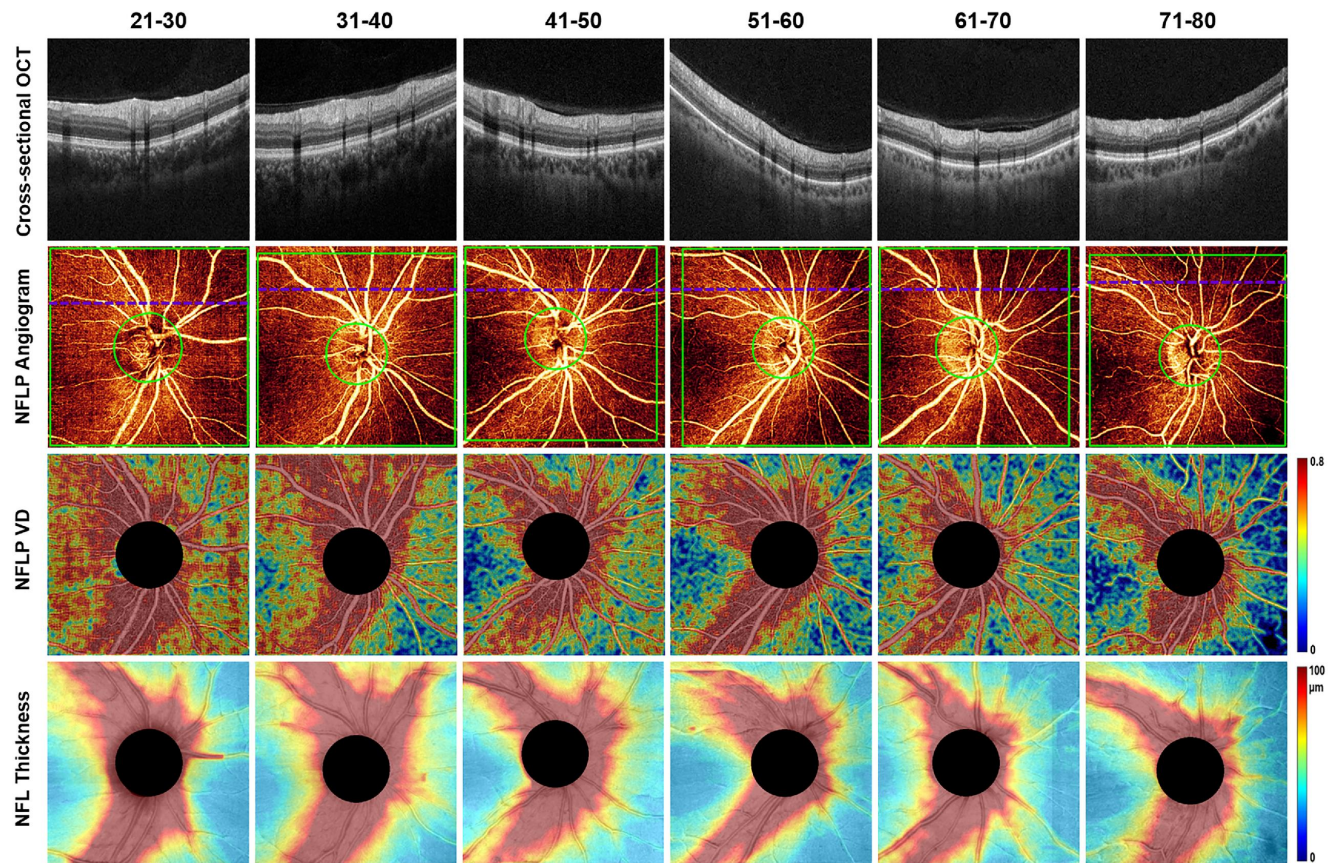


FIGURE 4. Representative OCT and OCTA images from different age-based subgroups of healthy controls. The cross-sectional OCT images (row 1) are shown with the corresponding location indicated by the *violet dashed line*. The NFLP angiogram (row 2) is overlaid with the measurement area, defined by the *green square* while excluding a 2-mm-diameter *circle* centered on the optic disc. The NFLP angiograms and NFLP vessel density (VD) heat maps (row 3) show a progressive reduction in VD with increasing age. NFL thickness (row 4) decreases in older age subgroups (> 50 years). The top number shows the age range of the subgroup. OCT, optical coherence tomography; OCTA, optical coherence tomography angiography; NFLP, nerve fiber layer plexus. NFLP VD, nerve fiber layer plexus vessel density; NFL, nerve fiber layer; NPDR, nonproliferative diabetic retinopathy; PDR, proliferative diabetic retinopathy.

and decreased NFLP VD. Because age showed stronger associated with reductions in both NFL thickness and NFLP VD (standardized $\beta = -0.25$ and -0.42 , respectively) than DR (standardized $\beta = -0.004$ and -0.26 , respectively) in our

cohort (see [Table 3](#)), this potential confounding effect should be considered when interpreting the relationship between DR severity and NFL structural and vascular alterations. To address this issue, we performed age-adjusted subgroup

analyses, in which no significant age differences were observed among the DR severity subgroups (see Supplementary Table S2). In the age-adjusted cohort, no significant differences in NFL thickness were found among DR severity groups and healthy controls, consistent with previous reports indicating that NFL thinning does not significantly vary across DR stages.^{29,30} Additionally, NFL thickness did not correlate with DR severity in this subgroup analysis. In contrast, a significant difference in NFLP VD was observed across different stages of DR (Fig. 1), and a significant correlation between NFLP VD and DR severity was noted, consistent with previous research.³¹ These findings from the age-adjusted cohort suggest that the observed reduction in NFLP VD is primarily associated with DR severity rather than age alone. The reduction in NFLP VD is likely due to microvascular dropout and impaired retinal perfusion, which are manifestations of DR.³² Together, these findings suggest that NFLP VD has the potential to serve as a biomarker for evaluating DR, whereas NFL thickness is less reliable for assessing DR progression. This also implies a disease mechanism in which perfusion is lost before atrophy or cell death.

There are several limitations in this study. First, its cross-sectional design limits our ability to assess longitudinal changes in NFL thickness and NFLP VD over time, restricting the evaluation of these biomarkers' utility for long-term monitoring of DR or AMD. Second, the relatively small sample size in advanced stages of the disease may limit the statistical power to detect subtle differences in group comparisons. Additionally, age was included as the primary covariate; however, other potential systemic and lifestyle-related factors were not evaluated and should be considered in future studies. Furthermore, this study included eyes with a history of intravitreal anti-VEGF injections prior to OCTA imaging. Although previous studies have generally reported no significant effect of anti-VEGF therapy on retinal NFL thickness,^{33–38} the potential residual effects of prior treatment on OCT/OCTA-derived NFL thickness and NFLP VD cannot be fully excluded.

CONCLUSIONS

This study quantitatively analyzed NFL thickness and NFLP VD using OCTA to evaluate their relationship with AMD and DR while accounting for potential confounding factors. Age was significantly associated with reductions in both NFL thickness and NFLP VD in eyes with AMD and DR. AMD was not independently associated with reductions in either parameter. In contrast, DR was significantly associated with decreased NFLP VD, but not with reduced NFL thickness. These findings suggest that reduced NFLP VD is associated with DR severity after adjusting for age and provide a basis for future studies investigating its role in DR.

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